

A Correlated Signals

Computational Game Theory

Model of Litigation Bargaining

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Abstract

A computational game theory algorithm solves extensive-form game trees representing litigation games in which parties receive correlated quality signals. Litigants make file, answer, agreement-to-bargain, offer, and quit decisions. The model allows examination of interactions among different game features and parameters. With baseline parameters, including risk neutrality, the parties are strikingly aggressive in their settlement demands, and settlement is rare. Fee shifting improves net recovery for meritorious plaintiffs and reduces net payout for meritorious defendants, while considerably increasing the excess burden above damages for truly liable defendants. This pattern extends somewhat to models with risk aversion, though less consistently.

1 Introduction

A natural assumption for settlement bargaining models is that the plaintiff and defendant receive correlated signals of the quality of the litigation. In a tort case, for example, each party may possess a combination of shared and private information about the events leading to an accident, and so each party can be understood as receiving a noisy signal of the true merits of the case. Even in a case in which facts are unambiguous and the only dispute is as to the law, each party will assess the case based on its lawyer's analysis and hunches, and the parties' estimates of case strength based on such legal advice will be correlated but not necessarily identical. Most of the literature modeling settlement bargaining, however, does not feature

correlated signals. In the classic models of Landes (1971) [15], Posner (1973) [19], and Gould (1973) [10], parties' beliefs about litigation quality are set exogenously rather than derived from signals. Later Bayesian models, such as Bebchuk (1984) [2] and Polinsky and Rubinfeld (1998) [18], assume that one party knows the quality of litigation while the other party knows the distribution of all cases but does not receive a signal of the particular case's quality.

Instead of correlated information, some models grant the plaintiff some information and the defendant some other information. The aggregate of these pieces of information then determines case quality. For example, Daughety and Reinganum (1994) [7] assign one party information on liability and another party information on damages. Similarly, in Friedman and Wittman (2006) [9], Dari-Mattiacci and Saraceno (2020) [5], and Abramowicz (2025) [1], the plaintiff and defendant each receive independent random signals from a known distribution, and the judgment depends on an average of those signals and, in the latter two papers, shared common knowledge. Thus, in these models, each of the plaintiff's and defendant's signals correlates with the merits, but unrealistically, the signals do not correlate with one another. The game thus becomes like a game of poker in which there are no common cards.

Plausibly, the optimal strategy and resulting equilibria in such a game might be quite different from those in a game of correlated signals. An intuition is that correlated signals would lead to high rates of settlement and relatively limited opportunities for strategic negotiation behavior. With independent signals, each litigating party must worry greatly about adverse selection; if the other party agrees to settle, that may well be because the other party has received a signal that the first party's claim is actually quite strong. But with correlated signals, there is only a small probability that the other party has a highly divergent view of the case strength. A reasonable strategy then might be to make demands or offers that rise roughly linearly with case strength, effectively agreeing roughly to share the surplus from settlement. Settlement might still fail because of tail estimates, but cases that do fail should resolve at roughly their expected value.

One pair of articles allows the plaintiff and defendant to estimate the quality of the litigation based on correlated signals. Lee and Klerman (2016) [16] and Klerman, Lee, and Liu (2018) [14] use correlated signals to update the canonical selection model of Priest and Klein (1984) [20]. In these updated models' benchmark, the true value of the litigation is Y (distributed uniformly over the entire real line under an improper prior), and the plaintiff and defendant receive signals of $Y + \epsilon_p$ and $Y + \epsilon_d$, respectively, where each ϵ is a normally distributed random variable. The papers, which also consider more general distributions, differ in that the older adopts the Landes-Posner-Gould assumption that the cases settle whenever there exists a settlement amount at which each party prefers to settle, while the more recent explicitly models the bargaining structure. One of the later paper's bargaining models (in addition to take-it-or-leave-it bargaining) is that of Chatterjee and Samuelson (1983) [4], in which each party separately makes an offer

and settlement occurs at the midpoint if the plaintiff's offer is less than or equal to the defendant's.

These papers represent significant advances in the realism of litigation models. But they have limitations that make it difficult to assess how correlated signals would impact issues other than the selection of cases for litigation. First, even in the more recent paper, trial-win probabilities are constrained relative to litigation costs so that each party may have a credible threat to take the case to trial (pp. 392-93). In effect, the plaintiff always files, the defendant always answers, and neither party ever quits before trial. Yet we have known since Shavell (1982) [21] and P'ng (1983) [17] about the importance of filing and answer decisions and since Huang (2004) [11] and Hubbard (2016) [12] about the strategic importance of the plaintiff's option to abandon a suit. Second, the entire cost of litigation is borne all at once, rather than sequentially. Yet expenditures on litigation are dispensed over time. Third, although case quality varies, there is a simple cutoff, above which the plaintiff has a high fixed probability of winning and below which the probability is fixed at a low value. More realistically, the probability of winning may be close to 50% in some cases and a range of high or low values in others.

In this article, I relax these assumptions through use of a computational model that identifies Nash equilibria in large extensive form game trees, applying the linear programming algorithm identified in von Stengel, van den Elzen, and Talman (2002) [23]. Abramowicz (2025) applies this algorithm to model settlement bargaining with game trees of up to 16,111 nodes but does not consider correlated signals of litigation quality. Rather, that article seeks to establish a proof of concept for computational (or algorithmic) game theory modeling by using a setup similar to that of Dari-Mattiacci and Saraceno, in which the parties have independent signals that aggregate to determine case quality. The baseline ten-signal, ten-offer specification in the present article has 47,611 nodes, while the eight-signal, fifteen-offer extension has 63,753 nodes.

Despite this greater complexity, the algorithm is designed to identify normal-form perfect Nash equilibria. The plaintiff and defendant each assign probabilities to the actions at each information set. The Nash criterion is that neither player could increase its expected utility, even by a small ϵ , by changing its strategy while holding the other player's strategy constant. Normal-form perfection requires each player's equilibrium strategy to remain a best response at every step of some sequence of opponent mixed strategies converging to the opponent's equilibrium strategy, with each element assigning positive probability to every pure strategy. The perfection requirement thus provides an equilibrium with a form of robustness to small opponent mistakes.

The primary solver uses rational arithmetic. Terminal utilities are discretized to integral values on a normalized range from 0 to 100,000, where the endpoints are the minimum and maximum achievable utility for that player in the game tree. (Setting the maximum at a higher value, so as 10^6 would add negligible additional precision, but at the cost of considerably more complex rational arithmetic.) Chance probabilities

are rounded to the nearest 10^{-5} and normalized to sum to 1. The equilibria are exact given these rounded utilities and chance probabilities.

The algorithmic game theory model thus follows in the Bayesian tradition of recent settlement bargaining scholarship, but with enhanced strategic complexity, including multiple levels of case quality, multiple signal levels, variable approaches to fee-shifting, adjustable levels of risk aversion, and litigant options to file, answer, and quit before trial. Though a computational model may not offer the elegance and generalizability of analytical models, the richness of game features can provide a useful complement to those models. The computational approach shines in three ways. First, it can illustrate how different game features may interact when analytical computation might be intractable. Second, the computational approach can make it possible to explore changes in rules, such as incorporating fee shifting, without developing an entirely new model. Third, welfare effects may be precisely measured, and different aspects of accuracy may be distinguished.

The evaluation of fee-shifting illustrates all of these advantages. Critical to assessment of the implications of fee-shifting is the choice of plaintiffs whether to file and of defendants whether to answer, as cases that end before settlement negotiations may represent either the optimal outcome from the system of litigation (the correct answer with no or minimal costs) or a miscarriage of justice (where the meritorious party declines to participate). Meanwhile, even if exit before trial is uncommon, a party that seems likely to exit may lose bargaining position. Interacting entrance, settlement, and exit is challenging, especially when layering on top of that fee shifting or risk aversion. It is no wonder that even a paper as sophisticated as Klerman, Lee, and Liu foregoes most of these features. Meanwhile, fee shifting can have multiple effects. Even separating litigation costs from accuracy can be crude, because both interact to affect the welfare of actual or potential litigants. Ideally, we would like to know, for example, how close meritorious plaintiffs come to being made whole, taking into account their litigation costs, or what the burden of litigation costs and judgments is on both truly liable and truly not liable defendants. The computational model allows these measures to be disentangled straightforwardly.

One of the most striking findings is that the parties are highly aggressive bargainers with risk neutrality, not so much deciding how much of the perceived settlement surplus they wish to insist on, but instead holding out for outcomes far more favorable than their assessments of case strength would suggest. Substantial bargaining failure occurs even though parties have relatively good information and even though the one-round structure eliminates opportunities to influence an opponent's offer through a bluff. Conditional on filing and answering, trial is a common occurrence with risk neutrality, even though nonfiling and nonanswering already culls the cases in which parties are least optimistic about their own cases. Risk aversion considerably reduces the incidence of trial, and the high rate of settlement under the American rule appears makes the risk averse specification appear to be a closer fit with our own observed patterns of litigation.

At the baseline parameter level with risk neutrality, the effect of changing from the American to the British rule is that nonfiling and nonanswering substitute for settlements. With both rules, when a case does reach the settlement stage, the defendant offers at most a nuisance settlement, while the plaintiff insists on strong damages, except under the American rule with a signal of 0.25, when the plaintiff accedes to a nuisance settlement. We do not see a schedule of offers that gradually increases with signals. With the British rule, a party with a favorable signal opts not to bargain at all. The additional difficulty in obtaining settlements under the British rule in turn explains why plaintiffs will not file and defendant will not answer under a broader signal range under the British rule. With both rules, parties effectively seek to hold out for highly advantageous settlements, but especially under the British rule, the signals that might lead an opponent to offer such a settlement instead lead the opponent not to contest litigation at all.

It might seem that this explanation is dangerously causal. Indeed, one must be careful in interpreting computational equilibria and especially in comparing different equilibria. This is not a just so story, however, but a set of conclusions that can be supported by considering how each party's best response would change when parts of the opponent's strategy are taken from one equilibrium and parts, from another. For example, we can detect that at one modestly unfavorable signal, the direct effect of fee shifting (that is, the effect holding the plaintiff's American rule strategy constant) is to increase answering, but the changes in the plaintiff's strategy under the British rule, mostly with respect to settlement behavior, push in the opposite direction, and the defendant answers less. An exact equilibrium represents many finely interacting decisions, such that changing any one might upset the balance, but the computational method allows us to examine how different parts of a party's strategy affect aspects of the other party's strategy, and thus one can at least assess which aspects of an opponent's strategy under a new equilibrium explain the other player's action probabilities at particular information sets.

Broadening across different levels of costs, the results cohere to tell a new and interesting story about fee shifting. Plaintiffs and defendants who are correct on the merits are better off with fee shifting than without, and the effect generally increases, the higher the level of costs for the relevant type of litigation. The average meritorious plaintiff, for example, will still not be made whole, considering both judgments and legal fees paid (and recovered), but the plaintiff does better with fee shifting. Meanwhile, the excess burden, if any, that a liable defendant must pay over and above damages, once taking into account fees and shifting, is much higher with fee shifting. These trends are clear under the risk neutral results even though overall measures of accuracy show essentially no effect. The effects are robust to risk aversion in most simulations, but there are some exceptions, such as a greater nonliable defendants' burden under risk aversion with high costs.

The same is true with respect to other robustness checks, such as by varying the noisiness of signals,

the timing of costs, when fee shifting is triggered, and the distribution of the true merits. Under many parameter specifications, in contrast to our risk neutral British rule example, settlement will occur in some cases, but the overall pattern concerning the British and American rule mostly holds. A multiple equilibrium robustness check using approximate equilibria usually confirms the results, though it shows that there are some equilibria (generally more fragile ones, susceptible to slight trembles in an opponent’s strategy) in which the results do not show.

The article proceeds as follows. Section 2 describes the information, litigation sequencing, and baseline parameter values for the model. Section 3 provides baseline results under both risk neutrality and risk aversion. It shows which changes in one party’s strategy explain the other party’s strategy changes in both the shift from the American rule to the British rule under risk neutrality or risk aversion, as well as the shift from risk neutrality to risk aversion under either rule. It also provides a welfare analysis across different cost levels for both risk neutrality and risk aversion. Section 4 provides tables showing how both the welfare measures and trial and settlement rates change when adding fee shifting under a range of model specifications. It also describes an attempt to identify multiple approximate equilibria at the baseline parameter values for both risk neutrality and risk aversion. Finally, Section 5 offers a caution conclusion, noting that although the data provide some support for fee-shifting, especially under conditions of risk neutrality, .

2 The Model

This part frames the correlated signals model in explicitly game theoretic terms by describing the extensive-form game tree. Section 2.1 describes the signals the parties and the court receive of the case quality. Section 2.2 sets out the sequence of litigation decisions and explains how the parties’ information changes as the game progresses. Section 2.3 specifies how litigation costs are accrued and assigned under the American and British rules and the trial-only extension, as well as how risk preferences affect utility. Section 2.4 summarizes model parameters and illustrates a path through the game tree. Section 2.5 defines social welfare outcome measures.

2.1 Information structure

The underlying strength of the plaintiff’s case, Q , is defined on a continuous range, distributed uniformly over $[0, 1]$ in the baseline simulations. Because the computational approach used here depends on extensive-form game trees, however, the game presented to the solver must have finitely many information sets and actions.

Let $\mathcal{S}_P$, $\mathcal{S}_D$, and $\mathcal{S}_C$ be the signal sets for the plaintiff, defendant, and court, and let $\mathcal{O}$ be each party’s offer set. The random signals X_P , X_D , and X_C take values in the corresponding sets. The baseline has

$|\mathcal{S}_P| = |\mathcal{S}_D| = 10$, $|\mathcal{S}_C| = 2$, and $|\mathcal{O}| = 10$.

A model with discrete signals and offers is admittedly less general in these respects than a model based on continuous quantities, and a virtue of the mathematical literature on settlement bargaining is that some variables can be modeled continuously. In Bebchuk (1984)[2], for example, the defendant knows the probability that the court will find liability, which can be any value from a mutually known probability density function. But not all variables in mathematical settlement models are continuous. In their review of the settlement bargaining literature, Daughety and Reinganum (2012) [6] include both two-types and continuous types models. Even models that offer continuous types in one respect feature only one or two types in another respect. For example, Klerman, Lee, and Liu (2018) [14] allow each party to obtain a continuous signal of the quality of litigation, but the objective probability of prevailing at trial can take on only one of two values. Thus, the conventions of the existing literature do not mandate continuous variables. The ten signal levels for each party allow a considerably greater degree of granularity than two-types models, while underlying case quality varies continuously, as in Klerman, Lee, and Liu.

The discretized signals of case quality that the plaintiff, defendant, and court receive are independent conditional on the realized value of Q . They are correlated signals, however, because each depends on Q . A party that receives a signal indicating that the plaintiff has a relatively strong case should conclude that the other party and the court will be more likely to also have such a signal or a nearby one. The level of noise is parameterized. Specifically, the parameters σ_P , σ_D , and σ_C represent the standard deviations of the normal noise used to generate the plaintiff's, defendant's, and court's signals. That is, for a given underlying case quality $Q = q$, each noisy assessment is drawn from a normal distribution centered on q , with the standard deviation specified for the party or court.

The interval $[0, 1]$ is divided into equal-width signal intervals, and the ten party-signal labels are the interval midpoints, from 0.05 to 0.95. Noisy assessments that lie outside $[0, 1]$ are excluded. Thus, with ten signal intervals, an assessment between 0.3 and 0.4 produces the signal labeled 0.35. Conditional on any q , the probability that a party receives a particular signal is the area under the normal density curve within that signal's interval divided by the total area between zero and one.

Because the underlying case quality is continuous, calculating the probabilities used in the game for discrete signals requires integrating over the possible values of Q . For $i \in \mathcal{S}_P$ and $j \in \mathcal{S}_D$, let $p_i(q) = \Pr(X_P = i \mid Q = q)$ and $d_j(q) = \Pr(X_D = j \mid Q = q)$ denote the corresponding conditional signal probabilities. The signals are independent conditional on q , so, under the baseline uniform distribution of quality,

$$\Pr(X_P = i, X_D = j) = \int_0^1 p_i(q) d_j(q) dq.$$

This integral accounts for every possible underlying quality that could produce the observed pair of signals. This makes it straightforward to calculate the probability of any defendant signal given a particular plaintiff signal. The integrals over case quality are evaluated numerically using 64-point Gauss–Legendre quadrature, which approximates each integral by a weighted sum of evaluations at selected quality values. For the baseline information structure, an automated test verifies that increasing the number of quadrature points from 64 to 128 changes no calculated signal or true-liability probability by more than 10^{-9} .

Figure 1 illustrates the relationship between case quality and the parties’ information. Because Q is defined continuously, the groupings on the left are illustrative only. The highlighted fan out demonstrates the relative probabilities that a plaintiff or a defendant will receive different signals given that the continuous case quality is somewhere between 0.6 and 0.8. The combined probability that the signal is within this range (either 0.65 or 0.75) is 39.9%, and the probability that the signal is less than 0.5 is only 17.9%. The informal intent is for information to be fairly strong. This baseline is selected because the paper focuses on correlated signals, and this approach addresses the concern that any strategic behavior that results might be attributable to the correlation being too low.

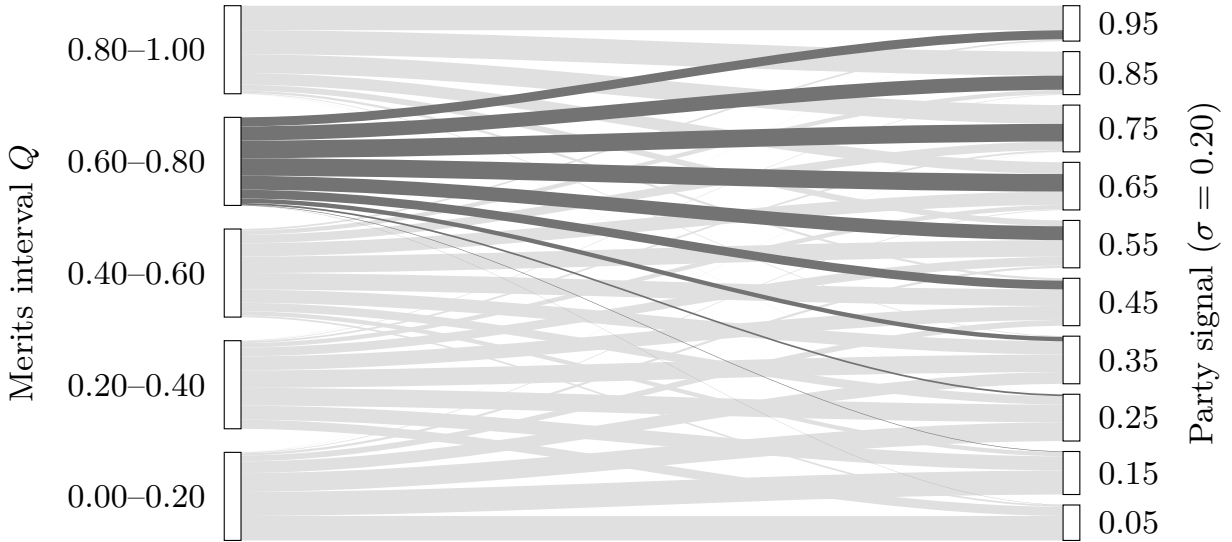


Figure 1: Information structure

A simpler design would restrict q to $\{0, 1\}$ and then to derive each party’s and the court’s signal directly from the true state. The approach of allowing a continuous Q is more consistent with the literature, particularly Klerman, Lee, and Liu, though in their benchmark they allow factual strength to vary across the entire real line while constraining trial-win probability to only two values. More significantly, continuous merits better exhibit an important feature of correlated signals, that a party expects that its opponent likely has

a signal close to the party’s own. Not necessarily so with a direct signal of truth. For example, with the symmetric signal distributions used here, when a party receives a signal of exactly 0.5 (as would be possible with an odd number of signals), the opponent is no more likely to receive a signal of 0.5 than if the party had drawn any other signal. The approach of drawing a signal directly from the underlying truth is, however, used as a robustness check in Section 4.1.

2.2 Litigation sequence and updating

The game begins with the plaintiff’s decision whether to file and, conditional on filing, the defendant’s decision whether to answer. Neither party knows its opponent’s signal. Although the game tree imposes an ordering, making it appear that the Chance decision determining the plaintiff’s signal occurs before the Chance decision determining the defendant’s signal, that ordering has no strategic effect. An information set groups decision points that a player cannot distinguish. When the plaintiff makes its decision whether to file, it arrives at an information set containing solely its signal. This may follow any of $|\mathcal{S}_D|$ defendant game nodes. The defendant observes filing and can infer information from it. At later stages, the parties condition on both filing and answering. Thus, their strategies reflect both their private signals and the information conveyed by participation, although no new evidentiary signal arrives during the game.

Assuming that the plaintiff files and the defendant answers, then the plaintiff faces a decision whether the plaintiff will abandon the litigation rather than incur trial costs if settlement fails, and the defendant similarly faces a decision whether to default in the event settlement fails. Neither party’s decision has any immediate effect, and neither party finds out about the other party’s decision. The placement of these decisions before settlement rather than after is admittedly counterintuitive. But it greatly speeds up the algorithm for calculating equilibria, relative to a model in which each player takes into account both its own settlement offer and its opponent’s in deciding whether to quit. With the latter approach, there would be one information set for each player for each triple of a liability signal, a settlement offer, and an opponent’s settlement offer. By deciding abandonment or defaulting in advance, the number of information sets in the game is greatly reduced, with just one information set concerning quitting for each liability signal for a player. This specification does not allow a party to revise its exit decision in response to the opponent’s observed offer. Including decisions whether a party will abandon or quit litigation continues to serve its primary role of affecting bargaining indirectly, by forcing each party to consider in making a settlement offer whether its opponent has a credible threat to take a case to trial.

The bargaining protocol includes two steps. First, each party decides whether it is willing to bargain at all. Second, if so, parties exercise the protocol introduced in Chatterjee and Samuelson (1983) [4] and used

in Friedman and Wittman [9], Dari-Mattiacci and Saraceno (2020) [5], and Abramowicz (2025) [1]. Each party submits an offer without observing the other's, and the case settles if and only if the defendant's offer is at least as high as the plaintiff's demand, with the settlement value equal to the midpoint. The offer actions are at the levels described in the prior section.

The reason for including the first stage is that if only the second stage is used, a party knows that it can at least obtain a nuisance settlement. For example, absent the first stage, a defendant with a strong signal of likely liability could make a settlement offer of 0.95 and then be assured of having that offer taken. That is problematic, because the worst case scenario for a defendant under fee shifting is worse than this, as the defendant might need to pay damages and both parties' legal fees. An alternative approach might be to add offer levels below zero and greater than one, but that would require either much larger game trees or reduced offer granularity. (Modestly increased offer granularity is considered as a robustness check in Section 4.) Moreover, even anecdotal evidence of such extreme settlements is rare. The agreement-to-bargain decisions maintain computational tractability while still eliminating any party's assurance at being able to achieve a settlement that, though highly favorable to the opponent, does not involve paying fees.

While future work might employ some other more complex bargaining protocol, such as back-and-forth offers between the parties, that would also increase the number of information sets and the computational burden. Chatterjee-Samuelson bargaining avoids the simplification in the Landes-Gould-Posner bargaining models of assuming that a case will settle whenever that is in the mutual interest of the parties, given their expectations. With Chatterjee-Samuelson bargaining, parties have some incentive to shade their offers in their own favor in the hope of capturing more of the settlement surplus, though they must balance this objective with the risk of bargaining failure. The approach thus allows for an important dimension of strategic bargaining, while avoiding the simplification adopted by models that arbitrarily choose a player to make a take-it-or-leave-it offer to the other player.

If a case does not settle but one party previously decided to quit in the event of settlement failure, the game ends without trial. If both parties decided to quit, a lottery assigns each a 50% chance of being the party that quits. This resolves mutual withdrawal once, rather than treating both parties as separately losing. If neither party quits after settlement failure, each incurs the additional trial cost and the court determines liability.

If bargaining fails and neither party quits, then a trial occurs. The court has two signal intervals, based on whether assessments lie below or above 0.5 and corresponding to no liability versus liability. Like the parties, the court can therefore make mistakes relative to true liability. The court's decision rule does not refine its inference based on the parties' filing, answering, or bargaining decisions. Making the court an additional strategic player that chooses a decision rule to optimize social welfare, more in the spirit of the

mechanism-design literature such as Spier (1994) [22], remains a possible extension.

2.3 Costs, preferences, and fee shifting rules

A plaintiff prevailing at trial receives a unit damages award. The allocation of costs depends on the fee shifting rule, and the parties' utilities depend on specification of risk preferences.

The ordinary-cost specifications set $c_{file} = c_{answer} = c_{trial} = 0.15$. Here c_{trial} is the additional trial cost for each party, so each spends 0.30 if the case reaches trial. There is no separate bargaining cost. Costs are fixed rather than chosen strategically. The model does not differentiate whether these costs take the form of attorney fees, court costs, or litigants' time. Each party's initial wealth is 10.0; with risk neutrality, this initial level does not affect the equilibrium.

A computational model also can easily consider the effects of risk aversion. Utility with constant absolute risk aversion is modeled as a function of wealth, $U = -e^{-\alpha W}$. The main risk-averse comparison sets $\alpha = 2$ for both parties. Utility values may be positively scaled and shifted without affecting the equilibrium. The Online Repository's Supplemental Materials folder contains a subfolder "Risk aversion utility curves" with a diagram illustrating utility as a function of wealth with this level of risk aversion.

Under the American rule, each party bears its own costs. The British rule requires that the loser pay all of the other party's costs incurred. This applies to a party that does not answer or to a party that exits before trial. This approach is generally similar to that of settled English cases, where each party bears its own costs. See, e.g., Civil Procedure Rules 1998, S.I. 1998/3132, rr. 12.5(2)–(3), 38.6(1), 45.16 (U.K.) (as amended). There are occasions in American law, however, where fee shifting is available unless a party exits early. See, e.g., Ariz. Rev. Stat. § 12-341.01 (restricting fee awards under the provision to contested contract actions); *Ungerer & Co. v. Alkaline88 LLC*, 2:23-cv-01153 (Feb. 16, 2024) (denying fees under that provision where the defendant never appeared or defended); see also Cal. Civil Code. § 1717 (barring contractual attorney fees on voluntarily dismissed contract claims). Meanwhile, some of the literature, including Farmer and Pecorino (2017) [8] and Huang (2004), limits fee shifting to occurring only after trial. Section 4 thus also considers a rule in which fee-shifting is triggered only by a loss at trial. In any event, no fee award attaches to nonfiling, and withdrawal still saves future trial expenses.

2.4 Parameter Values and Example Path

Table 1 summarizes the parameters of the model.

Table 1: Model primitives

Primitive	Value / interpretation
Merits and truth	Q is uniform on $[0,1]$; true liability $T \mid Q$ is Bernoulli(Q).
Signals	10 private signal bins per party; party noise 0.2; court noise 0.2; two court findings.
Damages and wealth	Damages 1; initial wealth 10 per party. Monetary quantities are in units of damages.
Preferences	Risk neutral or symmetric CARA risk aversion with $\alpha = 2$.
Costs	Filing / answering: 0.15 each; additional trial costs: 0.15 each; ordinary multiplier 1.
Private exit commitments	After filing and answering, each party commits whether to exit if bargaining fails.
Agreement and offers	Simultaneous agreement; offers only if both agree. Separate information sets; own commitment is remembered.
Offer actions	0.05, 0.15, 0.25, 0.35, 0.45, 0.55, 0.65, 0.75, 0.85, 0.95; overlapping offers settle at their midpoint.
Unsuccessful bargaining	Apply private exit commitments: abandonment, default, mutual-exit lottery, or trial. Refusal is not a terminal disposition.
Fee rules	American: own costs. British: loser pays at trial and the specified unilateral exits. Trial-only fee shifting is a separate cost-1 extension.

Figure 2 illustrates a portion of the game tree for the baseline simulation under the American rule. The C squares represent chance decisions, while the P and D circles represent plaintiff and defendant information sets numbered in the full game tree. Decision probabilities are rounded to three decimal places. The first panel illustrates two different possible signals, and those divergent paths are continued in the next two panels. The defendant does not know the plaintiff's signal. For example, at D_{23} , the defendant acts based on its anticipated utility absent this information, deciding to offer 0.05 because that maximizes utility. Meanwhile, P_{72} illustrates a mixed strategy, where the plaintiff is indifferent whether to file because each decision yields in expectation the same wealth and utility. The third panel in the diagram continues the first and illustrates two settlements, one on the equilibrium path and one off it, which both happen to occur when the plaintiff's demand equals the defendant's offer. Finally, trial probabilities are listed in the bottom panel.

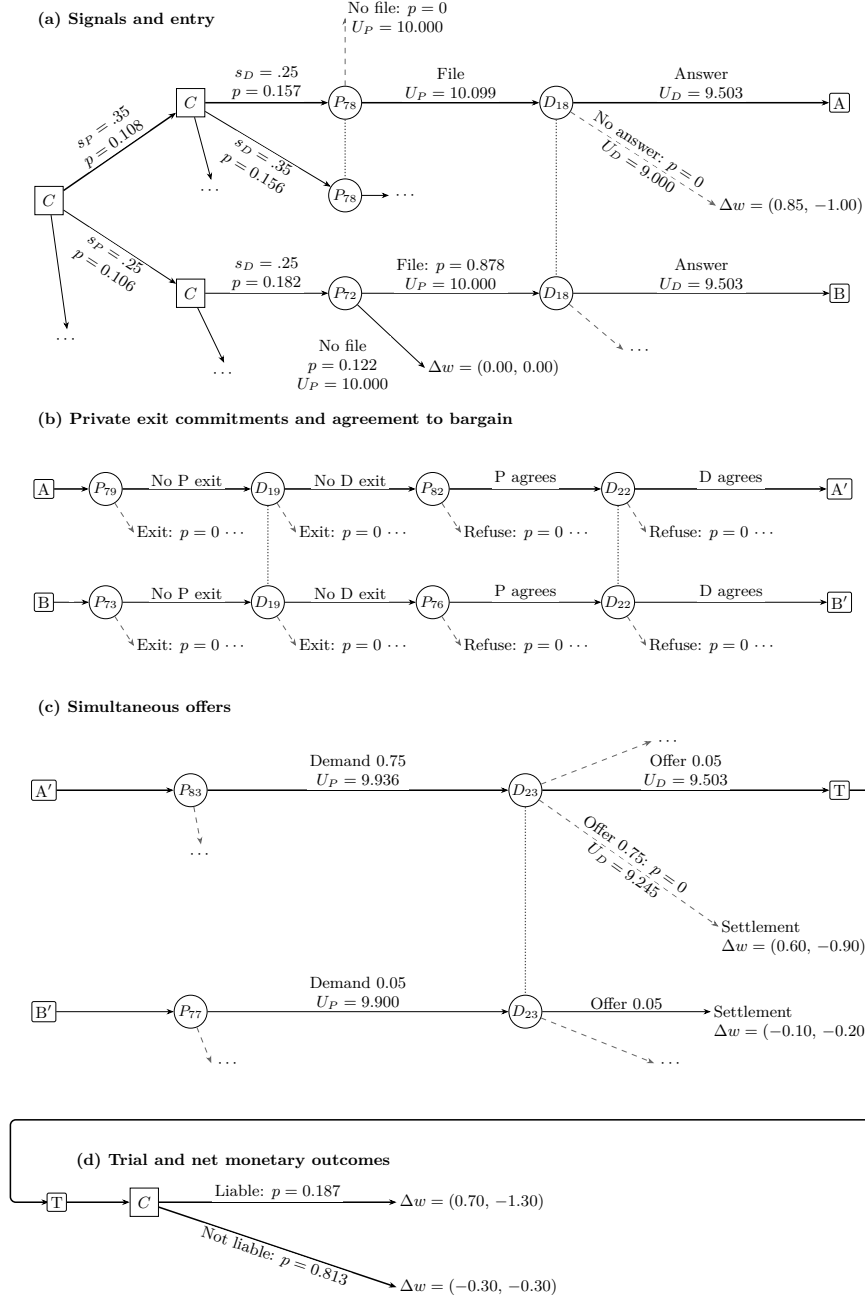


Figure 2: Worked equilibrium path

As a computational simplification with no effect on the equilibria, chance decisions at the end of games are collapsed. Replacing the court's decision with a single node reflecting the parties' expected utilities, taking into account risk aversion if applicable, leaves their incentives unchanged. Similarly, when both parties decide to quit, the expected utility from the mutual-withdrawal lottery can be calculated directly.

2.5 Outcome measures

Once an equilibrium is calculated, outcome probabilities and welfare measures can be evaluated by weighting the possible dispositions and underlying truth states. These are deterministic expectations rather than Monte Carlo estimates.

True liability T is binary, with $\Pr(T = 1 \mid Q = q) = q$.¹ Thus, although true liability and the strength of the plaintiff's case are correlated, a plaintiff can have a weak case despite true liability, or a strong case despite the absence of true liability. Neither party directly observes Q or T .

Explicit modeling of true liability permits measurement of the accuracy of outcomes. We can compare each party's net gain or loss with the outcome in a hypothetical perfect and costless litigation system. Defining accuracy solely by whether settlements mimic judgments would instead ignore imperfections in judgments themselves. A more ambitious approach would model the emergence of disputes endogenously in a particular legal context, as in Hylton (2002) [13], where a potential tort defendant initially decides how much care to take. This is left for a future extension, and the present model treats the population of potential disputes as given.

The monetary measures distinguish three interests. For a meritorious plaintiff, the shortfall is the amount by which net recovery, including litigation costs and fee transfers, falls below the unit damages award (or zero, if a full recovery is obtained). For a truly nonliable defendant, the burden is its net loss from litigation. For a truly liable defendant, the excess burden is its net loss above the unit damages award (or zero, if net loss is less than or equal to damages). Each measure is weighted by the probability of the relevant truth state, so that the reported figures are contributions per potential dispute, including disputes never filed.

Gross outcome error separately measures the absolute difference between the base payment and the payment warranted by true liability, before litigation costs and separately awarded fees. Real expenditures measure resources consumed in filing, answering, and trial. These measures distinguish errors in payments from the costs of obtaining a disposition. Their relative importance may be context-sensitive.

¹This can be derived by assuming that Chance first determines true liability, with $\Pr(T = 1) = \frac{1}{2}$, and then draws case quality from $Q \mid T = 1 \sim \text{Beta}(2, 1)$ or $Q \mid T = 0 \sim \text{Beta}(1, 2)$. These are probability distributions that have linear densities $2q$ and $2(1 - q)$, respectively. Their equal mixture is uniform on $[0, 1]$, and Bayes' rule yields $\Pr(T = 1 \mid Q = q) = q$. The Online Repository features extensions with different distributions of case merits.

3 Results

This part reports the principal results, focusing initially on the four baseline cases, involving both the American and British rules with both risk neutrality and risk aversion. Section 3.1 begins with the results under risk neutrality. In addition to showing the dispositions and the party strategies that produce them, it identifies the key explanations for the differences by examining best responses to opponents' equilibrium strategies under both rules. Section 3.2 similarly reports results for risk aversion. Section 3.3 considers the welfare implications of fee-shifting, extending the analysis to five different levels of cost, showing that fee shifting leads to higher welfare for meritorious parties, at least under risk neutrality, over all cost levels, while increasing the excess burden for truly liable defendants.

3.1 Risk Neutrality

Figure 3 reports the dispositions of potential disputes under the American and British rules with risk neutrality and ordinary costs. With the American rule, in 19.1% of potential cases, the plaintiff does not file, and in 37.3%, the defendant does not answer after the plaintiff files. A settlement occurs only 8.3% of the time, and the remaining 35.2% of potential cases go to trial, with the plaintiff losing more often than winning. Under the British rule, nonfiling and nonanswering increase, to 28.4% and 37.1% of potential cases, respectively, and settlement vanishes. It may be surprising that conditional on filing and answering, so few cases settle, though Chang and Klerman (2022) [3] do show that settlement rates vary greatly internationally. After all, the information structure was designed to give parties relatively good information about the strength of the plaintiff's claim, nonfiling and nonanswering provide additional information, and parties that settle can split the surplus from such settlement.

The most surprising aspect of the results is that no cases settle under the British rule. Of course, some cases do settle with fee-shifting in reality, leading to the question whether this is attributable to risk aversion, to players playing nonequilibrium strategies, or some other factor, such as the level of noise, the existence of litigants with different priors about the relevant signals. The point of the analysis, of course, is not to make a point estimate, and some settlement occurs in models only slightly different; for example, in a game tree with 12 signals per party and 8 offers (whose welfare results are summarized in Section 4), 4.2% of cases settle with the British rule. Whether settlement is rare or nonexistent, the results do reflect the conventional wisdom that the British rule may lead more cases to be resolved without litigation, as parties with relatively weak cases may be less likely to fight. In addition, the results reflect the familiar finding from Shavell (1982) [21, pp. 65–66] that fee shifting should make trial more likely conditional on filing and answering, because higher stakes should make mutually optimistic parties less willing to settle.

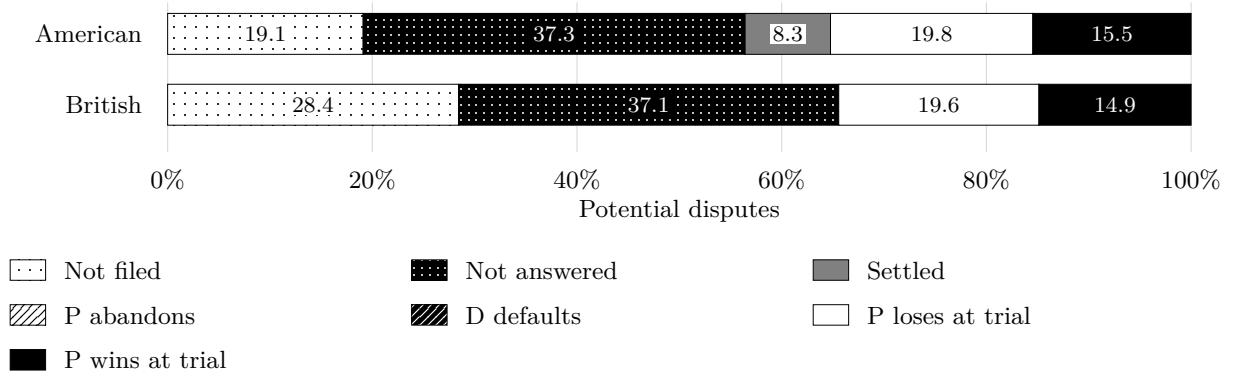


Figure 3: Risk-neutral dispositions

Figure 4 illustrates substantial strategic bargaining even with the baseline information structure. That strategic bargaining might occur is not surprising. All of the Bayesian models of settlement that use Chatterjee-Samuelson bargaining (including Friedman and Wittman (2006) [9]) recognize that each party faces a trade-off: Greater generosity increases the chance of avoiding trial, but greater stinginess increases the portion of the bargaining surplus that the offeror will receive if a settlement does occur. But strategic bargaining plays a considerably larger role than one might expect. Real-world bargaining includes many features not accounted for in this model, but this result tentatively suggests that litigants might rationally often fail to reach settlements even when their actual assessments of the litigation value are not far apart. At least, these results should make us cautious about accepting at face value the results of Landes-Posner-Gould models that assume settlement occurs whenever there is a settlement amount that both parties prefer to trial.

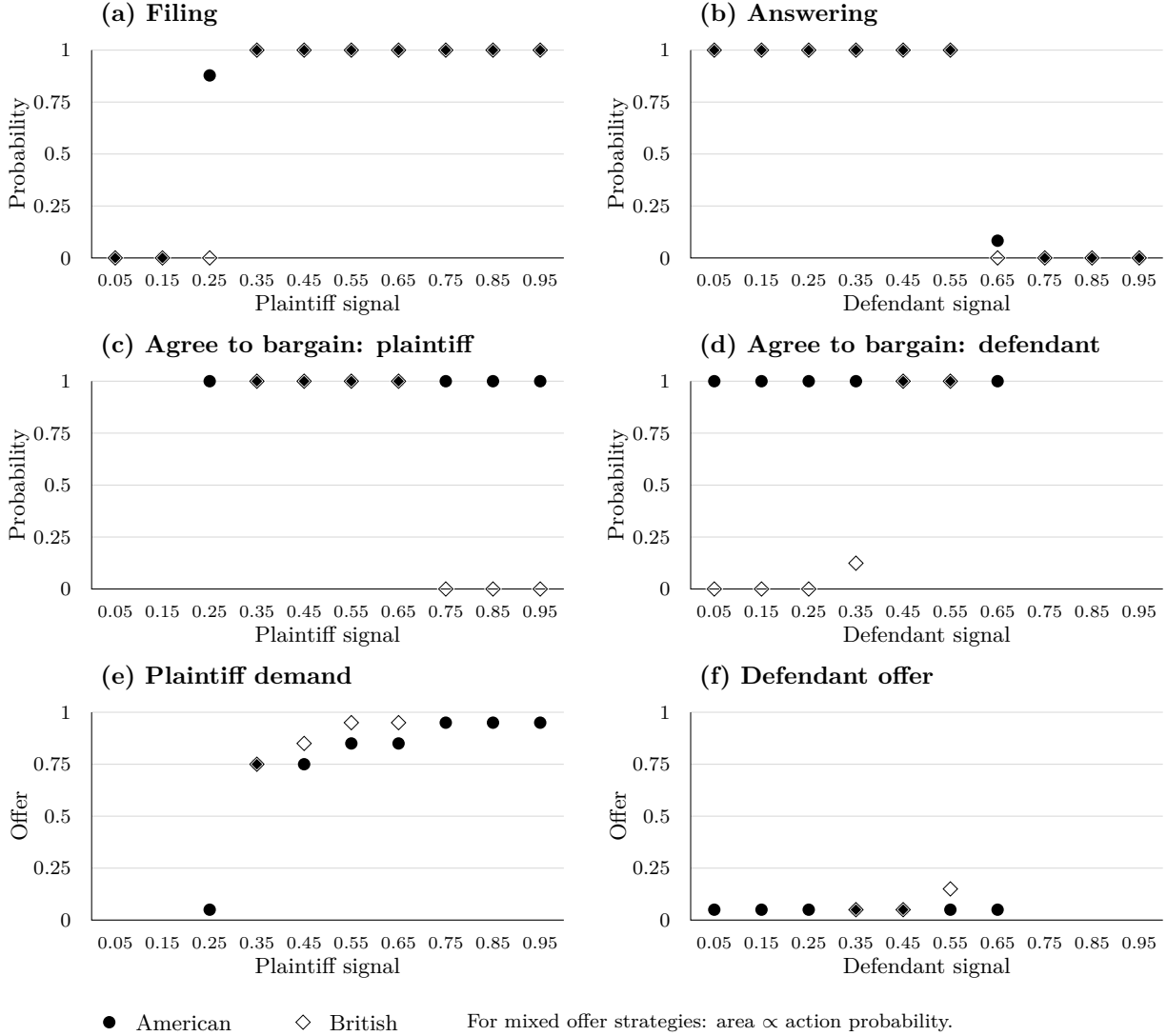


Figure 4: Risk-neutral participation and offers

As Figure 4 illustrates, parties do not file or answer when they receive the weakest possible signals, and nonfiling extends with the British rule to all cases in which the plaintiff receives a signal of 0.25. Meanwhile, the parties always agree to bargain with the American rule, but sense some benefit to refusing to bargain under the British rule, when they receive strong signal levels, given that they might win recovery of legal fees, including those already sunk. Possibly the most surprising aspect of the strategic bargaining is that the defendant never makes more than a nuisance offer of 0.05, except at one signal under the British rule when the defendant offers 0.15. In effect, the defendant under the American rule is holding out for the possibility that the plaintiff has received a signal of 0.25. Because the British rule strategies yield no settlements, it might appear that this is just an example of many equilibria, but that is not necessarily true, given that

the equilibrium be perfect requires that the strategies be at the limit of some set of opponent strategies including trembles, i.e. small mistakes. Still, we ought not place too much weight on the particular values, as the defendant will never come close to meeting the plaintiff demands for plaintiff signals greater than or equal to 0.35.

Table 2 helps explain the baseline patterns by identifying critical actions in the baseline strategies and disaggregating the interactions that led them to change or not. For example, we can see that the direct result of changing to the British rule without changing the defendant’s strategy from the American rule equilibrium is that it becomes modestly more profitable for the plaintiff to file at signal 0.25, yet the decline in the defendant’s willingness to bargain strongly counteracts this, and the probability that the plaintiff files at that signal falls from a probability of 87.8% to 0%. The defendant’s willingness to answer at another signal falls from just 8.3% to 0, yet here there is a much larger direct effect in favor of answering, and a range of changes in the plaintiff’s strategy that push in the opposite direction. The parties’ willingness to commit to exit and to agree to bargain, as well as offers, also reflect complex dynamics at various critical points.

Table 2: Selected risk-neutral strategy changes

American to British | risk neutral

Decision	Signal	Own exit	Action	Action %	Reach %	Direct	Entry	Offers	Exit	Agree
P files	0.25	–	Yes	87.8 → 0.0	10.62 → 10.62	+12.2	+0.0	+0.0	+0.0	-100.0
D answers	0.65	–	Yes	8.3 → 0.0	10.20 → 9.69	+91.7	-25.0	-41.7	+8.3	-41.7
P commits to exit	0.35	–	Yes	0.0 → 0.0	8.88 → 8.81	+0.0	+0.0	+0.0	+0.0	+0.0
D commits to exit	0.05	–	Yes	0.0 → 0.0	4.05 → 2.61	+0.0	+0.0	+0.0	+0.0	+0.0
P agrees	0.75	C	Yes	100.0 → 0.0	3.55 → 3.41	+0.0	+41.7	+8.3	-41.7	-8.3
D agrees	0.05	C	Yes	100.0 → 0.0	4.05 → 2.61	-100.0	+100.0	+0.0	+0.0	+0.0
P offer	0.45	C	0.75	100.0 → 0.0	7.70 → 3.18	+0.0	+50.0	-50.0	-33.3	+33.3
D offer	0.55	C	0.05	100.0 → 0.0	9.62 → 5.72	+0.0	+8.3	-58.3	+8.3	+41.7

Action/reach: percent. Contributions: percentage points. C/E: continue/exit.

Note: Direct changes the fee rule while holding the opponent’s strategy fixed. Opponent-entry and opponent-offer contributions average over the orders of replacing opponent strategy components. Filing and answering contributions are percentage points; demand contributions are fractions of damages. Opponent-exit and remaining contributions are zero in these rows. The change magnitude attributed to each cause can depend on the resolution of ties between equally good actions, as well as on the choices specified at unreached decision points. For example, an alternative specification of unreached choices would attribute only 66.67 percentage points of the filing decrease at signal 0.35 to answering, with the remaining 33.33 points split evenly between offers and exit. The total decrease remains unchanged. The full analysis and diagnostic checks are in the online repository.

3.2 Risk Aversion

Figure 5 reports dispositions when both parties have CARA risk aversion with coefficient $\alpha = 2$, holding ordinary costs and the information structure fixed. Figure 6 shows the corresponding participation and offer strategies. Table 3 presents the strategy changes discussed below, with the American rule in Panel A and the British rule in Panel B.

The dispositions are strikingly different for the American rule. The plaintiff always files, the defendant always answers, and settlement is reached 95.9% of the time, with trial occurring on the remaining occasions. Comparing the risk neutral to the risk averse equilibrium, as in Table 3, tells us that the increased willingness to file and answer arise entirely because of the changes in party offers. These offers now have considerable overlap, as shown in Figure 6, though still not exhibiting a gradual increase with increased signals. Rather, both are step functions, with offers rising from 0.25 to 0.75, albeit at different signals, reaching the higher value at 0.65 for the plaintiff's signal and 0.45 for the defendant's signal. Panel A of Table 3 attributes the selected offer changes to changes in the opponent's offers.

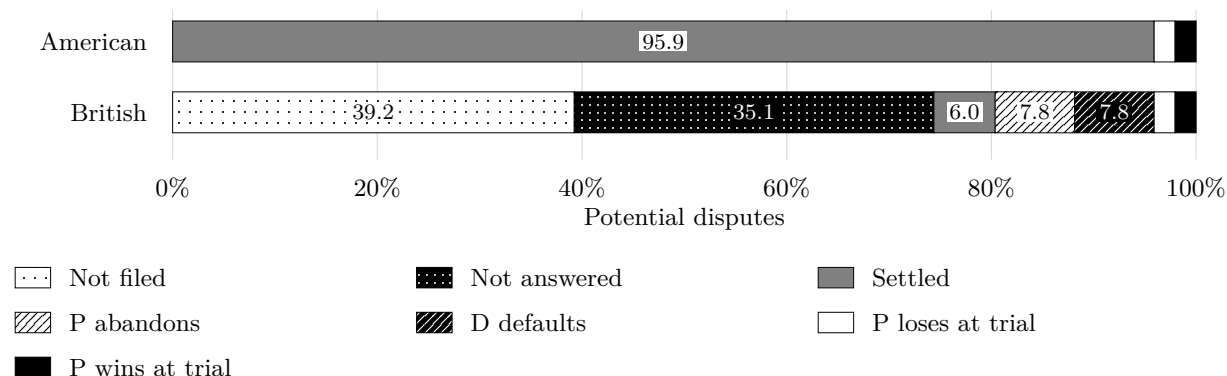


Figure 5: Risk-averse dispositions

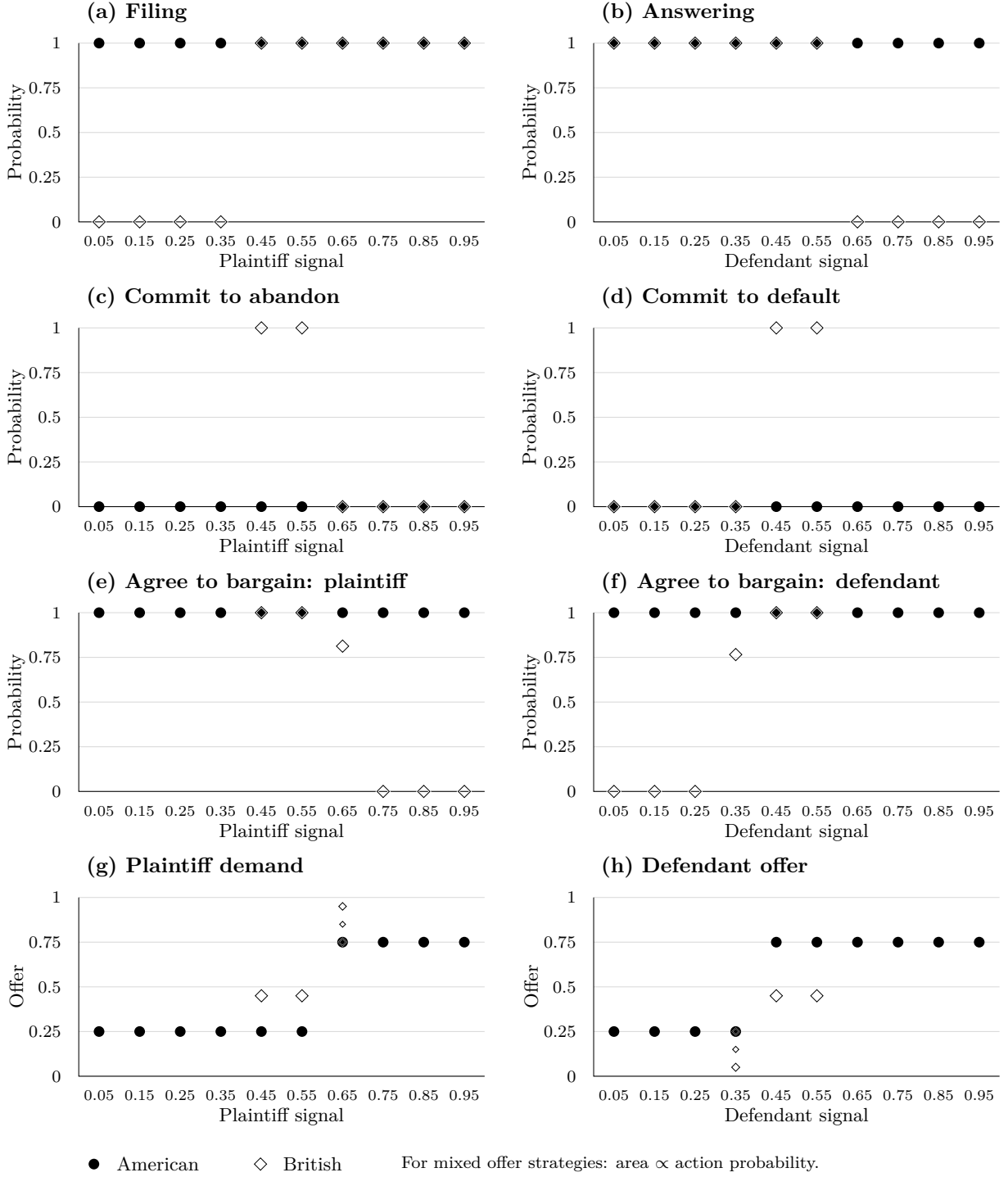


Figure 6: Risk-averse participation and offers

Table 3: Selected strategy changes from introducing risk aversion

Panel A. American | Risk neutral to risk averse

Decision	Signal	Own exit	Action	Action %	Reach %	Direct	Entry	Offers	Exit	Agree
P files	0.05	–	Yes	0.0 → 100.0	8.04 → 8.04	+0.0	+0.0	+100.0	+0.0	+0.0
D answers	0.75	–	Yes	0.0 → 100.0	10.33 → 10.62	+0.0	+0.0	+100.0	+0.0	+0.0
P offer	0.25	C	0.05	100.0 → 0.0	8.34 → 10.62	+0.0	+0.0	-100.0	+0.0	+0.0
D offer	0.05	C	0.05	100.0 → 0.0	4.05 → 8.05	+0.0	+0.0	-100.0	+0.0	+0.0

Action/reach: percent. Contributions: percentage points. C/E: continue/exit.

Panel B. British | Risk neutral to risk averse

Decision	Signal	Own exit	Action	Action %	Reach %	Direct	Entry	Offers	Exit	Agree
P commits to exit	0.45	–	Yes	0.0 → 100.0	7.60 → 7.60	+100.0	+0.0	+0.0	+0.0	+0.0
D commits to exit	0.45	–	Yes	0.0 → 100.0	7.67 → 6.16	+100.0	+33.3	+0.0	-66.7	+33.3
P files	0.35	–	Yes	100.0 → 0.0	10.81 → 10.81	-100.0	+0.0	+0.0	+0.0	+0.0
D agrees	0.35	C	Yes	12.4 → 76.6	6.41 → 4.72	+87.6	-29.2	+20.8	-79.2	+0.0

Action/reach: percent. Contributions: percentage points. C/E: continue/exit.

The British rule dispositions are closer to the results under risk neutrality. The plaintiff does not file in 39.2%, and the defendant does not answer in 35.1%, of all cases. This time, there is some settlement, 6.0% of all cases. Interestingly, the remaining cases include considerably more premature exits (7.8% per party) than trials. This reflects the simple intuition that risk-averse parties abhor trial. Panel B of Table 3 reflects that, while also showing that the plaintiff’s reduced willingness to file is also a direct effect of risk aversion. Some other changes in actions, such as the defendant’s willingness to agree to bargain at signal 0.35, conditional on a commitment to continue, reflect a complex mix of factors. All in all, the British rule results with risk aversion are broadly consistent with the observation that the British rule will lead to less litigation, though the data do not show it leads to any increase in trial.

3.3 Welfare Analysis

Figure 7 reports welfare measures under both risk neutrality and risk aversion, both at the baseline costs level and at other multipliers of this level. It compares monetary burdens, gross outcome error, and real expenditures across fee rules and risk preferences, using the definitions in Section 2.5. The first two columns of Figure 7 contain measures of the quality of the litigation system from the perspective of a meritorious plaintiff and of a truly not liable defendant, as any shortfall or burden in the net returns of these respective parties can be seen as an imperfection of the legal system. The third column measures the extent to which liable defendants bear an excess burden above damages. This is more difficult to interpret from a welfare perspective, as excess burden may represent either improved deterrence or overdeterrence, depending on the context. Either way, these three columns collectively reflect overall welfare effects in a way that takes into

account accuracy and litigation expenses. The fourth column shows the absolute error of judgments, ignoring litigation costs, but this is less useful because what matters to parties and may affect their behavior is not accuracy in the abstract but their net returns from litigation. Meanwhile, the last column illustrates the real costs of litigation, an important measure but one that is also already reflected in the first three columns.

The overall pattern is quite clear. With risk neutrality, fee shifting reduces the meritorious plaintiff's average shortfall and the nonliable defendant's average burden, with greater benefits generally arising when the overall litigation system is more expensive. For the liable defendant, however, the excess burden increases with fee shifting. These individual effects are not surprising; meritorious plaintiffs benefit from fee reimbursement or its anticipation, while nonliable defendants benefit from the reduced filing of nonmeritorious plaintiffs, and liable defendants suffer from having to pay opponents' fees. Note that gross outcome error reflects virtually no change in the aggregate, masking the combination of benefits for parties who justly should win and costs for defendants who should lose.

The pattern under risk aversion is largely the same, though a bit messier. Most notably, nonliable defendants benefit from the American rule rather than the British rule with high cost multipliers. The intuition is that when legal costs are very high, even plaintiffs with favorable signals may not file suit under the American rule, because the cost of litigating a case through trial may exceed damages. Such cases, however, become more feasible to bring with fee shifting, and even defendants who are not truly liable may ultimately settle, exit, or pay to defend themselves, and if the court errs, pay both damages and the plaintiff's fees.

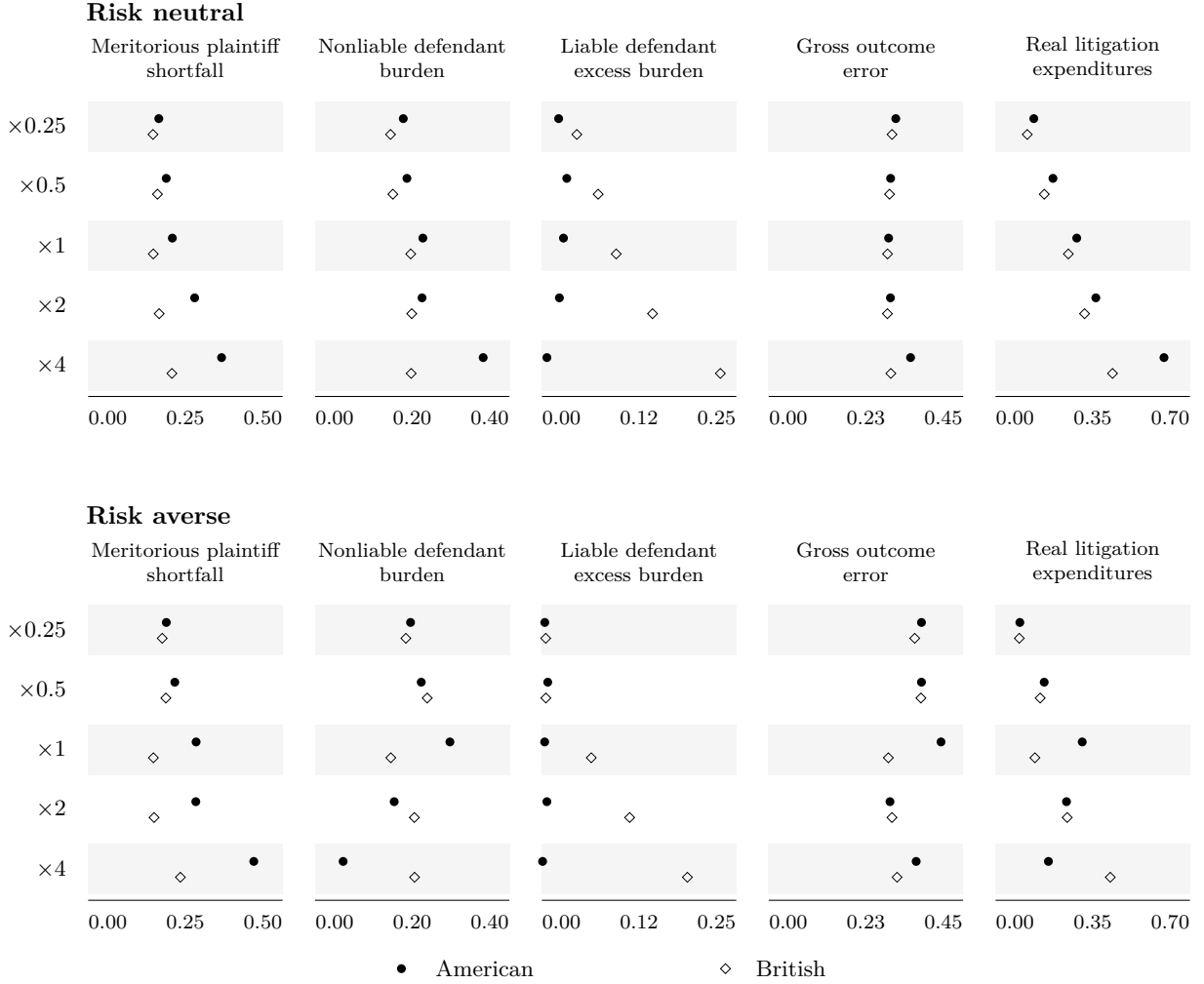


Figure 7: Welfare outcomes by cost multiplier under risk neutrality and risk aversion

4 Robustness

Robustness was assessed both by changing parameters and by searching for multiple equilibria. The Online Repository includes a Results folder that provides detailed information on each simulation, including data and generated figures, in addition to figures that aggregate and compare results across simulations. It also includes a Supplemental materials folder that includes equilibrium solution path animations, decompositions of changes in equilibria, game tree diagrams, diagrams illustrating different signals and their distributions with respect to one another and true merits, risk aversion curves, and multiple equilibria results.

4.1 Parameter Changes

To assess the robustness of the findings, various parameter values were changed.

First, the baseline cost levels were multiplied by each element of $\{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\}$.

Second, the timing of incursion of costs was varied. With later costs, $c_{file} = c_{answer} = 0.075$ and $c_{trial} = 0.225$. With earlier costs, $c_{file} = c_{answer} = 0.225$ and $c_{trial} = 0.075$.

[Update needed: Private-signal noise and court noise are varied separately in the executed cases; this sentence currently implies a joint variation.] Third, the parameters σ_P , σ_D , and σ_C were doubled to 0.4, for a high noise condition, and halved to 0.1, for a low noise condition.

[Update needed: Truth-conditioned latent merits is retired from the current collection. Calibrated direct binary uses the revised calibration target; distinguish merits variations from any still-undecided truth-map sensitivity.] Fourth, the distributions of truth and merits were varied. All specifications retain binary true liability $T \in \{0, 1\}$ with $\Pr(T = 1) = \frac{1}{2}$. In place of a uniform distribution of case merits, center-weighted merits and polarized merits assume that close cases are relatively common ($Q \sim \text{Beta}(2, 2)$) or uncommon ($Q \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$), respectively. Next, in the truth-conditioned latent merits model, the true liability state generates merits Q through a draw from $\mathcal{N}(T, 0.35^2)$ truncated to $[0, 1]$ and discretized into ten equal-width intervals represented by their midpoints. Finally, in the direct binary-state signals model, the parties and court receive signals directly of true liability T . The noise parameters are calibrated to approximate the joint distributions of the parties' and court's signals in the truth-conditioned latent-merits model.

[Update needed: The executed grids are 8/15, 12/8 and 8/8 signals/offers, not 10/15. Two British risk-averse rows are still pending in the recorded collection.] Fifth, the sensitivity of the offer grid was increased. The number of possible offer levels was increased from 10 to 15, while retaining ten signal levels for each party. The reason for focusing on offers rather than signals is the possibility that slight changes in the parties' offers could have large effects on the settlement rate. Finer-grained offers address this, at considerable computational expense, but do not prove convergence to a model with continuous offers.

[Update needed: Extensions now use cost multiplier 1 and American/British comparisons. The previous full crossing with five costs and three rules is no longer the run plan.] The noise, merits, and cost-timing alternatives were introduced without interacting them with one another. Each ten-offer specification was evaluated at all five cost multipliers under all three fee-shifting rules and both risk preferences. The fifteen-offer check was limited to the baseline cost level and other baseline parameters, again under all three fee-shifting rules and both risk preferences.

[Update needed: Table 5 now reports British-minus-American differences, not sign counts]

over 46 settings. Replace this interpretation and its universal-answering claims; two grid contrasts remain pending.] Table 5 provides a summary of how some key outcome variables compared across fee-shifting rules over all of the robustness checks. The individual cells indicate the number of simulations in which the outcome variable for the first listed fee-shifting rule was greater than, equal to, or less than the same variable for the second listed fee-shifting rule. Each comparison covers 46 settings per risk preference, using a tolerance of 10^{-5} for ties. Filing and answering shares are measured over all potential disputes; answering share is the joint probability that the plaintiff files and the defendant answers. Several regularities are notable. First, in every simulation, defendants always answered conditional on plaintiffs' filing under complete fee-shifting. Second, complete fee shifting usually produced no more filing than the other rules, though there were exceptions, particularly under risk aversion. Third, gross outcome error and meritorious-plaintiff shortfall were no higher with trial than under complete fee-shifting in all but two simulations in each of the risk neutral and risk averse conditions. Under risk neutrality, nonliable-defendant burden was more often higher under trial than complete fee-shifting, while liable-defendant excess burden was always at least as high under complete as under trial fee-shifting. Both comparisons were more mixed under risk aversion. Fourth, compared with the American rule, trial fee-shifting generally reduced meritorious-plaintiff shortfall under both risk preferences. Both fee-shifting rules generally reduced nonliable-defendant burden, especially under risk neutrality, but complete fee-shifting had mixed effects on meritorious-plaintiff shortfall. Both rules increased liable-defendant excess burden in most risk-neutral simulations, with more exceptions under risk aversion.

Table 5: Outcome differences: British minus American

Risk neutral British minus American							
Specification	P shortfall	D nonliable	D excess	Gross error	Real costs	Trial (pp)	Settle (pp)
Costs: earlier costs	-0.061	-0.029	+0.083	+0.000	-0.055	-8.4	+0.0
Costs: later costs	-0.056	-0.031	+0.027	-0.007	-0.093	-14.4	-20.7
Trial-only fee shifting	-0.004	-0.047	+0.020	-0.005	-0.057	-9.6	-8.3
Private noise 0.1	-0.050	+0.010	+0.064	+0.000	-0.000	+0.0	+0.0
Private noise 0.4	-0.051	-0.078	+0.063	+0.002	-0.103	-17.0	-0.4
Court noise 0.1	-0.050	-0.038	+0.060	-0.001	-0.071	-3.1	-18.3
Court noise 0.4	-0.048	-0.118	+0.045	-0.055	-0.175	-29.8	+1.3
Grid: 8 signals / 15 offers	-0.054	-0.058	+0.064	-0.018	-0.094	+0.0	-36.8
Grid: 12 signals / 8 offers	-0.063	-0.009	+0.058	-0.001	-0.038	-5.8	-3.8
Grid: 8 signals / 8 offers	-0.041	-0.047	+0.064	-0.002	-0.073	-11.7	+0.0
Merits: center weighted	-0.063	-0.060	+0.072	-0.031	-0.108	+0.0	-47.5
Merits: polarized	-0.053	-0.018	+0.068	-0.000	-0.025	+0.0	-7.7
Calibrated direct binary	-0.087	-0.059	+0.070	-0.003	-0.061	-8.1	-13.6

Table 5: Outcome differences: British minus American (continued)

Symmetric risk aversion | British minus American

Specification	P shortfall	D nonliable	D excess	Gross error	Real costs	Trial (pp)	Settle (pp)
Costs: earlier costs	+0.023	-0.041	+0.010	+0.005	-0.015	-7.7	+16.0
Costs: later costs	-0.077	-0.084	+0.037	-0.116	-0.056	+0.0	-80.7
Trial-only fee shifting	-0.082	-0.093	+0.010	-0.119	-0.138	-0.8	-84.8
Private noise 0.1	-0.088	-0.116	+0.054	-0.076	-0.217	-0.8	-96.1
Private noise 0.4	+0.050	-0.050	+0.000	-0.000	-0.000	+0.0	+0.0
Court noise 0.1	-0.087	-0.099	+0.053	-0.064	-0.194	-7.8	-82.1
Court noise 0.4	+0.001	+0.001	-0.000	+0.003	-0.002	-0.6	+0.6
Grid: 8 signals / 15 offers				Pending			
Grid: 12 signals / 8 offers				Pending			
Grid: 8 signals / 8 offers	-0.026	-0.143	+0.004	-0.108	-0.109	+0.3	-47.4
Merits: center weighted	+0.021	-0.010	-0.004	+0.018	-0.014	-4.6	+4.6
Merits: polarized	-0.108	-0.114	+0.059	-0.090	-0.205	-5.7	-88.1
Calibrated direct binary	-0.081	-0.081	+0.008	-0.151	-0.000	+0.0	-4.0

Table 5: Outcome differences: British minus American (continued)

Asymmetric risk aversion | British minus American

Specification	P shortfall	D nonliable	D excess	Gross error	Real costs	Trial (pp)	Settle (pp)
Plaintiff-only risk aversion	+0.024	-0.063	-0.003	+0.015	-0.095	-16.8	-1.2
Defendant-only risk aversion	-0.104	+0.045	+0.053	+0.010	-0.057	-13.8	-3.2

4.2 Multiple Equilibria

Finding an equilibrium does not establish whether it is representative of other equilibria of the same game. The computational approach therefore raises the question whether the results depend on which equilibrium the algorithm finds. I address this by varying the initial strategy profiles supplied to the solver and examining both the distinct profiles recovered and the variation in their outcomes. The search requested 50 starting profiles for each of the four American/British and risk-preference combinations (200 starts), of which 199 passed the approximate acceptance criteria. This exercise cannot determine with certainty how many equilibria exist. In principle, the recovered profiles could be the entire population or a small sample. *[Update needed: These are obsolete search/group counts. The current four-case search accepted 50, 50, 50 and 49 starts; distinct-profile counts require an explicit grouping criterion and tolerance.]* The exercise produced 21, 11, and 15 profiles for the three fee-shifting rules under risk neutrality, and 12, 19, and 7 under risk aversion. The counts distinguish complete strategies, which can differ only at unreached decision points and still produce the same outcomes.

[Update needed: Figure 8 now shows welfare outcomes rather than disposition ranges. The current risk-neutral searches show outcome variation. These are accepted approximate profiles, not a proof of distinct exact equilibria or their likelihood.] As shown in Figure 8, the multiple equilibria under risk neutrality all produced the same dispositions breakdown within each fee-shifting rule. Under risk aversion and the American rule, settlement ranged from 13.8% to 95.9%. Some risk-averse equilibria produced lower expenditures or lower gross outcome error than the risk-neutral equilibrium; expenditures ranged from 0.182 to 0.336, compared to 0.292 under risk neutrality, and the gross outcome range was 0.276 to 0.399, compared to 0.278. The risk-averse burden ranges under different fee-shifting rules also overlapped.

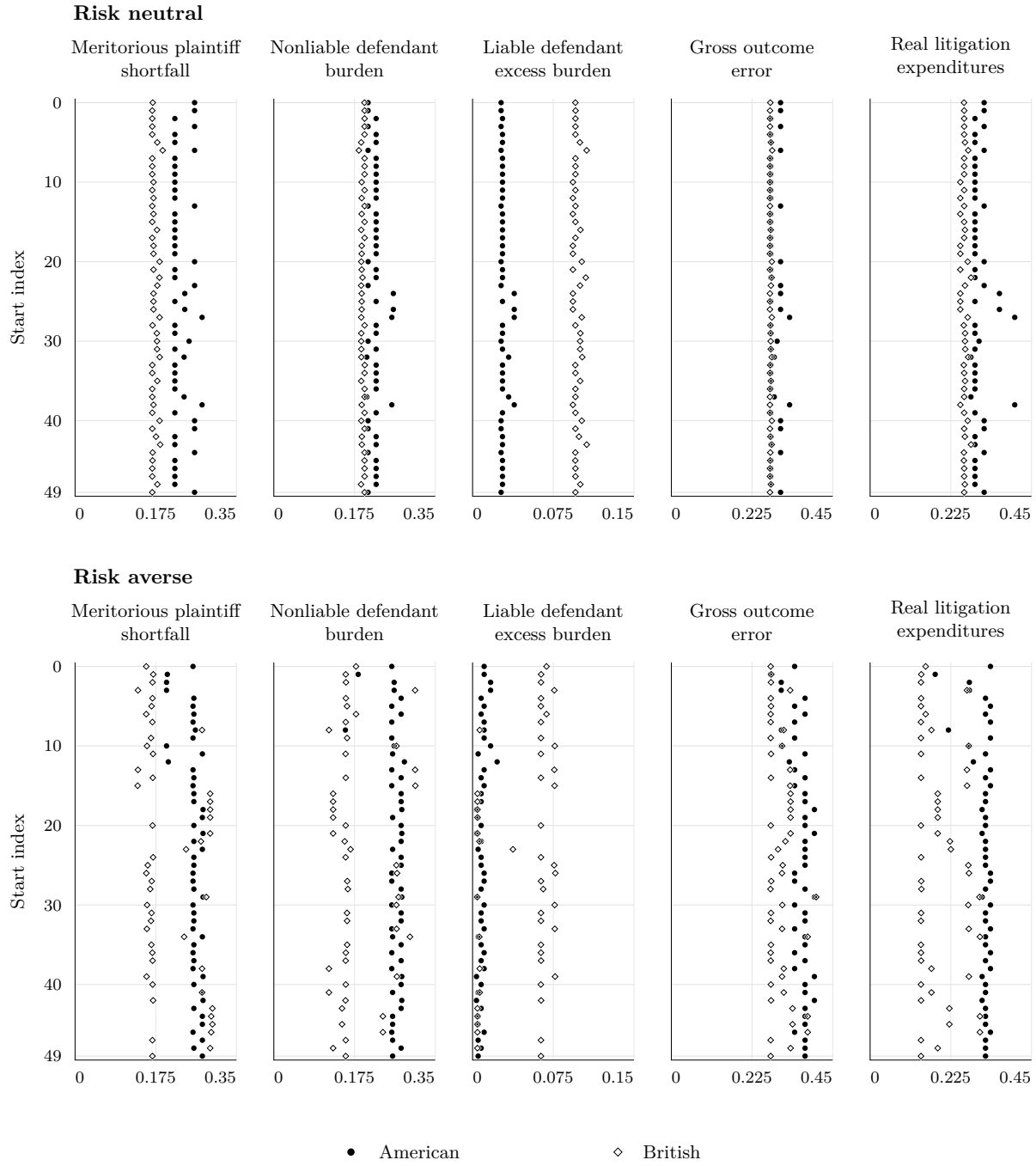


Figure 8: Multiple equilibrium welfare outcomes

5 Conclusion

[Update needed: The concluding universal-answering mechanism and trial-only welfare ranking are superseded. Preserve the general framing but reassess those conclusions using current results.] The computational model developed here complements analytical models by allowing interaction of different model features and variation of parameter values. The results do not produce an unequivocal win for either the American rule or the English rule writ large, but do show how changing what triggers fee-shifting may have significant effects on outcomes. In particular, with complete fee-shifting, defendants' incentive is to always answer, and this inevitability will in equilibrium tend to discourage plaintiffs with borderline signals from filing. In the baseline comparisons and most tested settings, a fee-shifting rule that applies only after trial produces a smaller expected monetary shortfall for meritorious plaintiffs than the more expansive rule, while the analysis from the perspective of the burden on a nonliable defendant is much more equivocal.

At least three limitations affect the interpretation of the model. First, litigants may not be rational Bayesians who play equilibrium strategies. But, as the literature generally presumes, models assuming rationality may provide a first guess at how litigants behave, and economic forces may gradually push actors in any given legal context toward equilibrium strategies.

Second, models' usefulness may depend on parameter values, and the parameters here are not calibrated based on data, either statistics from actual litigation or on microfoundations of individual behavior. A future direction would be to calibrate parameter values based on data for particular types of litigation in particular jurisdictions.

Third, the model omits important features of litigation, including learning and bargaining over time, as well as the role of lawyers and agency costs. Indeed, this is but a first step toward building models that can incorporate even more complex features of the litigation game. Because the game tree size may increase exponentially with additional features, other computational game theory approaches that seek approximate equilibria may be needed to make larger game models solvable.

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